1. Introduction

In the development of large language models (LLMs), a critical point of focus is the model's ability to interpret or understand the prompt. We define a model's *understanding* as its ability to correctly identify and preserve the intended meaning of the prompt when producing an output, independent of whether the model can correctly solve the task specified by the prompt. In other words, it is possible for a model to *understand* the prompt, yet still possibly produce an incorrect answer. While an incorrect answer is still undesirable, a more critical flaw arises if the model produces an output that derives from a flawed so-called understanding of the prompt, which may lead to an output inconsistent with the user's intent.

For example, suppose an LLM is provided with the prompt, "List the United Nations member states in alphabetical order." If the model successfully *understood* the prompt, it may possibly begin listing various states which may or may not be a United Nations member state. If the model did not *understand* the prompt, it may output unrelated information, such as a list of animals.

It may be difficult to distinguish between a *misunderstanding* and an incorrect answer. For the purposes of this study, both of these errors will be categorized as a *misunderstanding*.

This study will focus on how text deletions in textual input affect LLM sentiment classification accuracy. Deviations from the user's original purpose may potentially appear when the model processes modified or incomplete textual input if semantic analysis proves insufficient or inaccurate, leading to inefficiency in the prompting process since more context is then required to produce a satisfactory result. In other words, the language used by a human may differ from the LLM's understanding of language, leading to the model *misunderstanding* the user's intent, which, at best, wastes additional information to reorient the model or, at worst, produces an erroneous result. In particular, it is possible for such models to fail on the more subtle nuances of language, such as emphasis based on not just the words used but also the position of words in text. Stated generally, the primary issue is LLMs producing unreliable subjective text analyses due to a failure to properly interpret context and information contained within the text.

LLMs have become increasingly prevalent in modern society. For example, the research community has seen recent examples of applications in a variety of fields, such as medical research and mathematics. Accordingly, developers of LLMs bear greater responsibility in ensuring the accuracy and efficiency of the relevant models. In these situations, a misguided output may prove more disastrous than a failure to output anything at all. Thus, it is crucial for LLMs to be able to precisely *understand* the prompt given by the user, considering all potential facets of language.

Prior studies on LLM sentiment analysis have been performed, primarily focused on the derivation of sentiment from semantics and connotations within the language. In order to look at the issue from a different perspective, this study analyzes *degraded text*, referring to text that has had parts entirely deleted. By observing LLM sentiment analysis performance on *degraded*

text, it is possible to isolate the effect of certain types of degradation or deletion on text sentiment.

Subjective texts present a particularly useful setting for studying potential misinterpretation because their intended meaning often depends on sentiment, emphasis, and contextual relationships between words. As such, in order to properly address this problem, subjective texts must be substantially considered, since potential issues in more objective subjects may first appear in a subjective situation. Inferred knowledge from subjective data may also potentially prove insightful in a study geared towards objective data due to the underlying similarities by virtue of common language structures. Therefore, this study will orient its goal towards subjective texts specifically.

This study attempts to provide insight into how the structure of subjective human text is interpreted by LLMs, utilizing positive and negative movie reviews as a testbed. Specifically, it examines how the removal of information from these texts affects an LLM's ability to accurately determine their sentiment. It does not attempt to measure internal representations or establish a reasoning process within an LLM, as these topics are beyond the scope of this study. The hope is that the data provided by this study will assist developers of LLMs by exposing potentially hidden details in LLMs understanding of text. Subjective texts in particular are prone to this failure, as such texts rely significantly on human sentiment to communicate their ideas.

The research conducted in this study ultimately culminates in a single optimal text deletion policy which aims to preserve sentiment in subjective texts. This final result will deepen understanding of how sentiment is communicated and expressed, providing a foundation for further research on this topic.

2. Related Work

This study aims to analyze the semantics in language, specifically the English language, by analyzing subjective texts. The methodology and concepts utilized draws inspiration from several prior works primarily geared towards analyzing LLMs.

The idea of using movie review as a dataset is drawn by Maas et. al. in their work studying vector embeddings for words utilizing sentiment analysis. The study introduced the idea of using IMDb movie reviews as a dataset for sentiment analysis. By combining unsupervised and supervised learning, Maas et. al were able to encode both semantics and sentiment in the vector representation. However, they did not touch upon the degradation of prompt contents, which will be the primary focus of this study.

After Mass et. al. set the precedent, research on sentiment analysis with LLMs often used the IMDb dataset. These other studies share many similarities with this study, but leave behind certain gaps that must be addressed.

The methodology utilized in this study focuses on the impact of noise and compression on the understanding of LLMs. In a study analyzing natural language attacks on machine learning algorithms, Jin et. al. introduced TextFooler, a baseline for generating adversarial text aimed to reduce classification accuracy. The research evaluated how such attacks can severely impact model accuracy and understanding. Additionally, Herrera et. al. discussed in a recent study how prompt-level noise can impact LLM performance and accuracy and proposed a transformation layer that cleans input text prior to LLM inference.

The idea of prompt compression also appears in this study. Similar works include Jiang et. al. 's research, which produced LLMLingua. This prompt compression method identifies less important tokens in the prompt, removing them to eliminate redundant information while preserving optimal LLM performance. Many other studies have been built upon the foundation created by LLMLingua, including LLMLingua-2 by Pan et. al., which trained a model to determine which tokens should be removed, treating it as a token-classification problem; and LongLLMLingua by Jiang et. al., focused on removing entire sentences and passages rather than just tokens.

While extensive study has already been done on model accuracy with the interference of prompt noise, there is minimal research on the impact of information loss rather than corruption. By isolating this topic, this study focuses on the impact of text structure on understanding rather than simply semantics. Additionally, while LLMLingua and other similar prompt compression methods typically operate on individual tokens or sections of text, this study uses deletion policies based on text position, independent of semantics. As such, this research will primarily focus on the impact of text structure and placement on the perceived importance of tokens in the context of sentiment analysis.

3. Vision and Research Objective

3.1. Vision

The long term vision of this research is to provide a foundation for studying LLM *understanding* via analysis of incomplete data. By performing basic diagnostic research on how certain deletions applied on subjective text have varying effects on *understanding,* the study analyzes how LLM sentiment-classification reacts to changes in text structure. The optimal text deletion policy presents a clear report on how sentiment is maintained even through intense degradation. The data provided by this may potentially allow deeper insight into sentiment analysis and merely contribute towards improvements in new LLM architecture, which will not be discussed in this study.

3.2. Research Questions

This study will address three research questions in the context of subjective texts.

- RQ1: How does LLM accuracy change as deletion severity increases?
- RQ2: Does the method of deletion have significantly differing effects on accuracy?
- RQ3: Based on the gathered knowledge, is it possible to design a deletion policy to preserve the most information and, consequently, result in a consistently higher accuracy compared to other methods?

The scope of this study does not extend to the creation of a new architectural model in order to improve accuracy. The study will only contribute empirical data derived from existing models.

3.3. Hypotheses

A number of hypotheses are proposed for each research question:

- H1.1: Sentiment-classification accuracy will strictly decrease as deletion severity increases.
- H1.2: The rate of degradation will vary across deletion policies.
- H2.1: Deletion policies that preserve the structural information most relevant to sentiment classification will produce higher accuracy than random deletion at equivalent deletion severities.
- H3.1: A structure-informed deletion policy can consistently outperform other methods when applied to the analyzed data (movie reviews).
- H3.2: A structure-informed deletion policy can consistently outperform other methods when applied to other data not previously analyzed.

4. Methodology

The data collection process is facilitated with a Python script, utilizing the OpenAI library and the GPT 4.1 Mini model to perform sentiment-classification. Data visualizations are provided by the Python Matplotlib library.

4.1. Study Design

The experimental factors are the deletion policy used and the *severity* of the deletion. Each deletion policy and *severity* is used once on each movie review considered. This study uses 5 deletion policies and 9 *severity* levels.

The degraded review, stored in the Python variable *text*, is then fed into a LLM with the following prompt:

“Classify this movie review as Positive or Negative.
Review: {text}
Answer with only one word: Positive or Negative.”

The LLM's sentiment analysis is then compared with correct sentiment, according to the IMDb dataset, in order to determine the accuracy.

Every movie review considered is used once for each deletion policy/*severity* combination. In other words, every combination is operating on the same set of reviews, in order to minimize the possibility of experimental variance.

4.2. Dataset

This study uses the IMDb movie review dataset, a popular option for binary sentiment analysis. The dataset contains 50000 polarized reviews: 25000 for training and 25000 for testing. Negative reviews defined with a rating of 4/10 or lower and positive reviews defined with a rating of 7/10 or higher.

For this study, a total of 500 reviews are used for the initial data collections, while a further XXX are used to evaluate the human-created optimal deletion policy. These reviews are selected from the 25000 training reviews using simple random sampling.
The 25000 movie reviews in the training set are evenly divided in terms of positive and negative sentiment, with 12500 positive and 12500 negative reviews. Due to the methodology used in this study, the review lengths are measured in model tokens. The review lengths ranged from as little as 8 tokens to as much as 2883 tokens. The mean length is 285.43 tokens, the median length is 213.50, and the standard deviation of the length is 213.53.

4.3. Deletion Policies

Five deletion policies will be analyzed, as well as applicable combinations of policies:

- Random Deletion: a random selection of tokens are deleted. This provides a baseline to compare all other deletion policies to.
- Frontal Deletion: tokens at the front of the prompt are prioritized for deletion.
- Rear Deletion: tokens at the back of the prompt are prioritized for deletion.
- Sentence-based Frontal Deletion: tokens at the front of sentences are prioritized for deletion.
- Sentence-based Rear Deletion: tokens at the back of the prompt are prioritized for deletion.

At all times, the deletion policies will maintain a certain proportion of tokens in the modified output based on the *severity*. This study uses *severities* ranging from 0.0 to 0.8, taking measurements at each tenth. The proportion of deleted tokens compared to total tokens is roughly the related *severity* for all policies.

Random Deletion utilizes simple random sampling to delete tokens until the related *severity* is met.

Frontal and Rear Deletion deletes a certain proportion of tokens, corresponding to the *severity,* at the front and rear of the text, respectively. In other words, a token will not be deleted unless all other tokens prior to or following it, respectively, in the original text are also deleted.

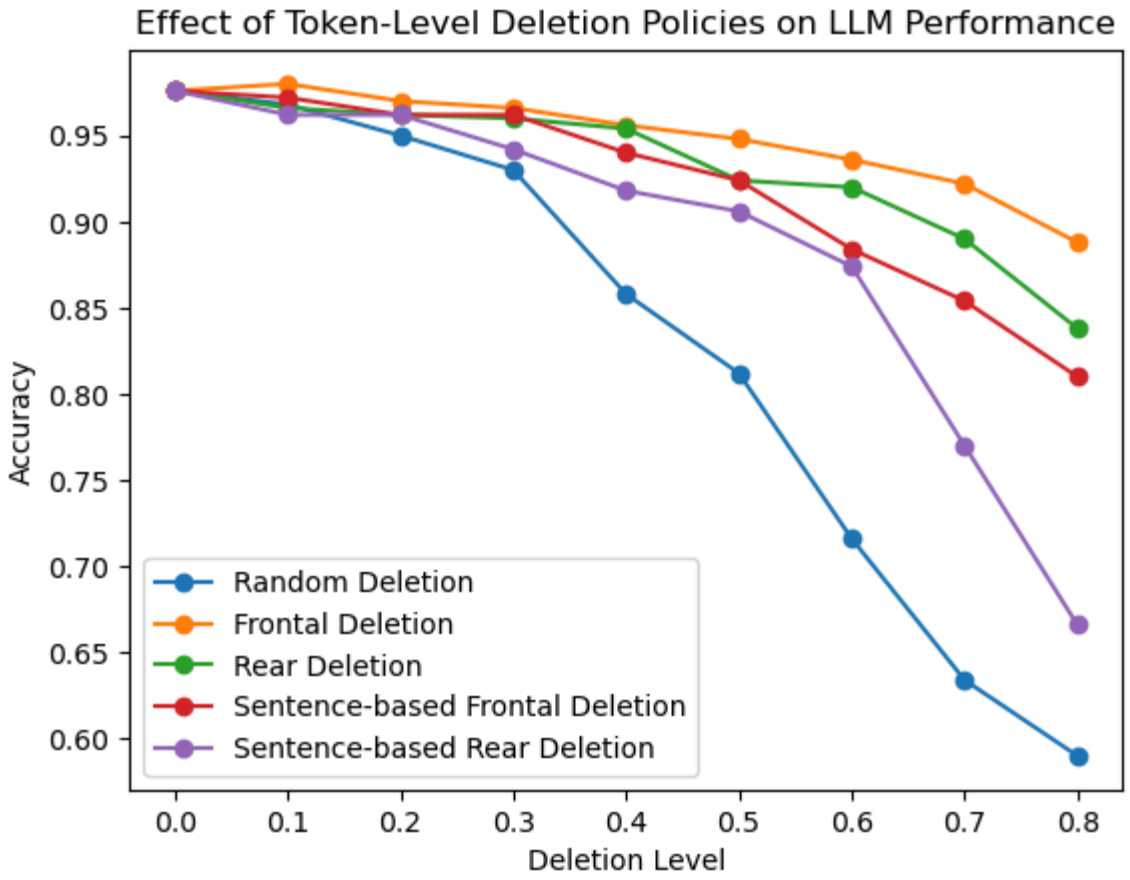
Sentence-based Frontal and Rear Deletion deletes a certain proportion of tokens, corresponding to the severity, at the front and rear of sentences and text. In other words, a token will not be deleted unless all other tokens prior to or following it, respectively, in its sentence of the original text are also deleted.

5. Results

5.1. Data

The results of the initial data collection are displayed below, in terms of accuracy rate:

Severity	Random	Frontal	Rear	SB Frontal	SB Rear
0.0	0.976	0.976	0.976	0.976	0.976
0.1	0.968	0.980	0.966	0.972	0.962
0.2	0.950	0.970	0.962	0.962	0.962
0.3	0.930	0.966	0.960	0.962	0.942
0.4	0.858	0.956	0.954	0.940	0.918
0.5	0.812	0.948	0.924	0.924	0.906
0.6	0.716	0.936	0.920	0.884	0.874
0.7	0.634	0.922	0.890	0.854	0.770
0.8	0.590	0.888	0.838	0.810	0.666

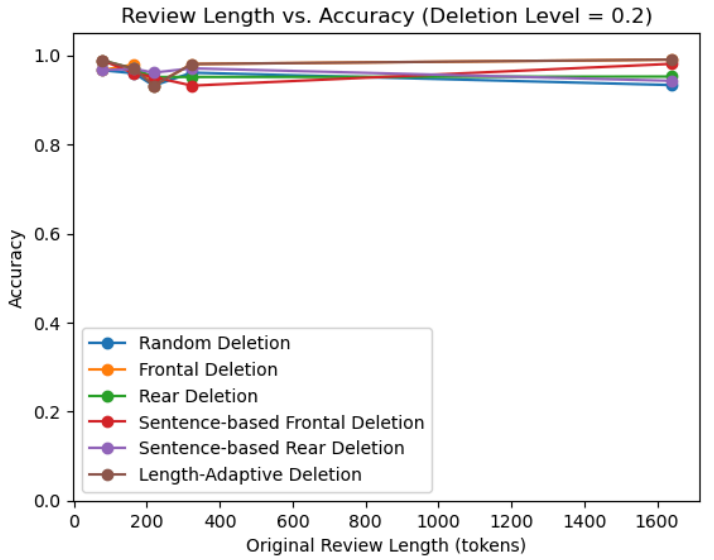


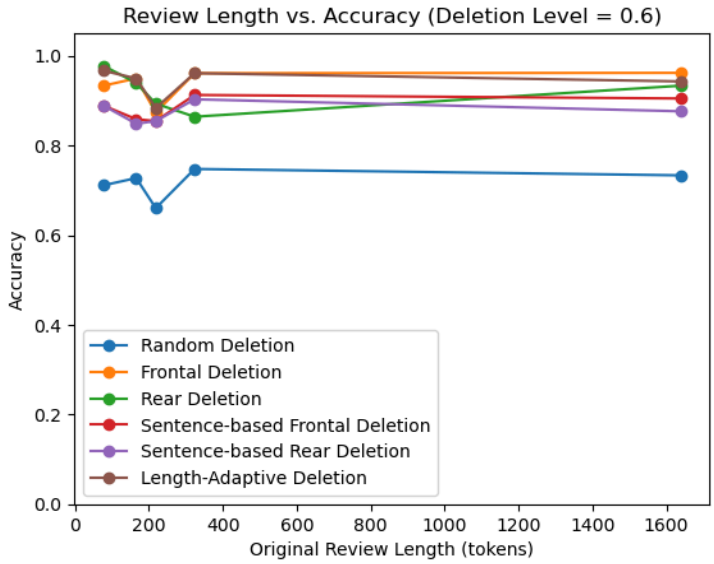
As seen above, the deviation in accuracy between policies in *severities* 0.0 and 0.1 seem minimal. Likely, there hasn't been enough information loss from any of the policies to prove significant to the LLM's *understanding*.

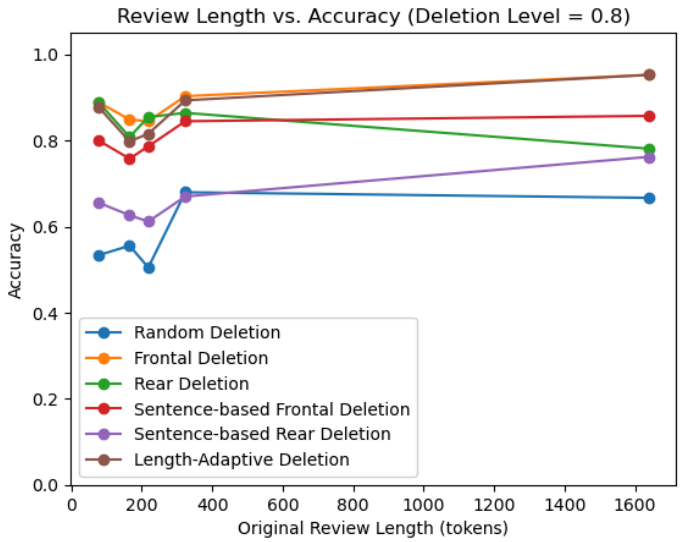
At 0.2 and 0.3, we can see trends begin appearing, with Random Deletion clearly experiencing the greatest decrease in accuracy. Random Deletion experiences a significant drop between 0.3 and 0.4, then holds roughly the same downward trend throughout. Sentence-based Rear Deletion experienced a significant drop from 0.6 to 0.7, with a similar drop from 0.7 to 0.8. The other three policies seem to hold a slightly parabolic or exponential course in their accuracy decreases.

We also note that Frontal Deletion consistently led to the least decrease in accuracy, experiencing no notable changes in its trend. In contrast, Rear Deletion and Sentence-based Frontal Deletion seem to oscillate slightly at lower *severity* levels, before stabilizing into recognizable trends at around 0.6.

Several notable plots relating accuracy and review length are shown below.







Note that we are primarily focused on differences in accuracy between policies rather than differences in accuracy between review length.

An interesting pattern we see in Sentence-based Frontal Deletion and Rear Deletion often swap placements as review length increases. At lower and higher *severities*, Rear Deletion tends to perform better initially but fall short with longer reviews, while the opposite happens with intermediate *severities.*

5.1. Optimal Deletion Policy

Work in Progress (working to find one)

5. Discussion

Work in Progress (cannot continue until optimal deletion policy is found)